

Exercise 03: Deque

A deque (a double-ended queue, pronounced “deck”) permits insertions and removals at either end of its collection. This lets it function as either:

- a **stack**, which follows the FILO (First-In, Last-Out) principle — the newest item added is the first one removed, or
- a **queue**, which follows the FIFO (First-In, First-Out) principle — the oldest item added is the first one removed.

All deque operations can be implemented with constant time complexity, $O(1)$. Write a C++ program named ‘main03.cpp’ that satisfies the following requirements:

1. Include only the libraries ‘`iostream`’ and ‘`string`’.
2. Define the `insertFirst()` function,

```
void insertFirst(string& s, char c)
```

that concatenates `c` to the front of `s`.

3. Define the `insertLast()` function,

```
void insertLast(string& s, char c)
```

that concatenates `c` to the back of `s`.

4. Define the `removeFirst()` function,

```
void removeFirst(string& s)
```

that removes the first character from `s` if `s` is not empty; otherwise, it does nothing.

5. Define the `removeLast()` function,

```
void removeLast(string& s)
```

that removes the last character from `s` if `s` is not empty; otherwise, it does nothing.

6. Testing in the `main()` function:

- a. Initialize a string variable to the lowercase alphabet.
- b. Invoke each of the four functions above at least once, on that same string object, in any order, choosing your own character(s) to insert. Before and after each function call, print the current value of the string.
- c. Verify that the functions work with the empty string.